

Project: LLM-Driven Database and Query Generation

Real database traces and queries are hard to get because of privacy and access limits. To work around this, we want to generate synthetic benchmarks that closely mimic how real databases and queries behave. Existing workload generators in this space (built on things like Redset and Snowset) can't achieve plan-level matching, because the fields those datasets carry (number of joins, bytes scanned, cpu time) say nothing about how a query's operators are actually connected or how many rows flow through each one, which is what a query plan actually is.

Our approach takes a logical plan directly as input: its operators (scan, filter, join, group-by), how they're connected, and how many rows go in and out of each one - the similar type of information already visible in a captured EXPLAIN ANALYZE. Our goal is to generate SQL text and schema for that structure, and a separate data generator for building a database, engineered so its real row counts satisfy every operator's target simultaneously. Efficiency of the system is measured by running the generated query on the generated database on the same engine as ground truth workload, and checking whether the resulting plan faithfully mimics the execution of ground truth.

Input side (logical plan)

